

Research Statements

The research statements afford you the opportunity to articulate your academic plans in relation to the objectives of the DOE CSGF. Reviewed in the context of your complete application, your responses shed important light into your way of thinking as influenced by your experiences and goals.

Field of Interest and the Role of Computational Science

- a. In terms a general audience would understand, describe an important, outstanding scientific or engineering challenge in your field of interest where computational science can play an important role.*
- b. Describe the particular science or engineering problem that you would like to pursue in your research and how you would approach this. What would be the impact on the field and/or on science, engineering and/or society in general if this challenge could be successfully addressed?*

Vehicle-to-Everything (V2X) communication is a rapidly expanding technology at the center of smart city transportation systems. Recent work at Argonne National Laboratory demonstrated that traffic infrastructure in a real urban corridor, beyond simulation, can communicate directly with vehicles to exchange information such as location, signal timing, speed, and braking status [1]. By integrating V2X enabled roadside infrastructure deployed by the Chicago DOT with a production electric vehicle, this experimental work demonstrated that simulated traffic models can be validated and have the capability to be meaningfully integrated into real-world environments. During my internship at Argonne, I helped demonstrate how hardware-in-the-loop testing and these traffic simulation models can bridge the gap between simulation and deployment in smart cities.

A central challenge to this research is how to scale such systems across dense urban environments, where traffic behavior is shaped by sensor data and by human-driven, real-world events that are not explicitly captured through existing infrastructure alone. Social networks generate vast, real-time data streams that reflect how people experience and report physical events such as crashes, road closures, and congestion. I propose modeling these online interactions as a single component of a broader cyber-physical systems, using them to complement V2X sensor data in identifying traffic anomalies that may not yet be visible or readily anticipated at individual intersections.

Computational science plays a central role in sustaining this research vision. Edge-based systems can detect localized anomalies (i.e., isolated crash in one intersection), but collaborative edge intelligence allows connected intersections to coordinate and share information across a city. Building on prior work in HPC-based traffic simulation, I aim to explore how large-scale, HPC models can integrate sensor data and social data to support real-time traffic control and safer urban mobility.

Citations (Optional)

[1] Jihun Han, Miriam Di Russo, Debashis Das, Qian Peng, Jongryeol Jeong, Michael Pam-minger, Priyash Misra, Dominik Karbowski, Kevin M. Stutenberg, and Trevor Alexander Crain. Advancing connected vehicle capabilities with scalable and flexible anything-in-the-loop testing framework. In 2025 IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+EATS), pages 1--6, 2025.

[2] Ugirumurera, J., Severino, J., Wang, Q., Sorensen, H., Todd, A., Nag, A., Potter, K., Jones, W., Ravulaparthi, S., Berres, A., Nugent, P., Moore, A., Tennille, S., Peterson, S., Xu, H., & Sanyal, J. (2019). High Performance Computing Traffic Simulations for Real-Time Traffic Control of Mobility in Chattanooga Region. <https://www.nrel.gov/docs/fy20osti/75009.pdf>

Use of Computational Science in your research

- a. *What is the most complex calculation you have run on a high-performance machine as part of your research experience? Or if you haven't run a high-performance computing system, tell us about the most complex computational problem you have tackled.*
- b. *Imagine if you were given access to resources 100 times more powerful than what you have access to. What would that enable you to do, and what do you perceive the mathematical and computer science challenges to be?*

The most complex computation I have conducted was training a cross-embodiment visuomotor behavior cloning system that jointly models human and robot action sequences from multimodal data. In this project, human demonstrations were first processed using 3D hand pose reconstruction to estimate temporarily aligned hand trajectories from video. On the other hand, robot demonstrations captured high-frequency joint states, Cartesian end-effector positions, gripper states, and camera-relative spatial transformations at each timestep. This task was computationally complex due to temporal alignment across modalities sampled at different frequencies, learning long-horizon dependencies over high-dimensional sequences, and scaling training to hundreds of thousands of optimization steps. Training runs that I led typically reached upwards of 600,000 optimization steps on our local HPC cluster using GPU hardware, with careful tuning of batch size and data normalization. In parallel, I led the collection of both human and robot demonstrations, acquiring 500 episodes per modality across four manipulation tasks. Compiling and preprocessing this multimodal dataset, including images, robot joint states, and human hand pose estimates, required distributed processing across multiple nodes and GPU resources to efficiently prepare the data for training.

If I were given access to computational resources 100 times more powerful than what I had access to, I would focus on scaling both the diversity and duration of embodied learning experiments rather than solely increasing model size. In the context of robot learning from human demonstrations, this would enable training on more diverse human video datasets, longer temporal horizons and action-chunking, and a wider range of embodiments and tasks.

From a mathematical perspective, the primary challenge would involve understanding how representation learning behaves under extreme data heterogeneity and long-horizon sequence modeling, with a particular focus on stability, convergence, noise, and uncertainty. From a computer science standpoint, scaling these experiments introduces significant challenges in distributed training, data synchronization, fault tolerance, and efficient training.

Citations (Optional)

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Program of Study

Describe how the courses listed in your planned program of study would help prepare you to address the challenges you have described in questions 1 and 2. Discuss your rationale for choosing these courses.

Addressing traffic incident detection and response through social media and Vehicle-to-Everything (V2X) data is fundamentally focused on addressing a socio-technical cyber-physical systems challenge that is multi-faceted and requires careful consideration of analyses, implementation, and scalability.

Distributing smart-city sensor infrastructure introduces complexities in network security, latency, and systems coordination, especially when modeling these interactions in real-time and in dense urban environments. Through my program of study at Purdue University, I aim to focus on computer and communication methods for product control, given that these approaches directly inform the coordination, synchronization, and visualization of vehicles, infrastructure, and other environmental factors within large-scale cyber-physical systems. V2X data embedded within smart city infrastructure relies on data communication frameworks to securely and reliably relay critical information across networks, highlighting the necessity to apply data communication and computer networks and distributed systems to this work.

V2X data can provide detailed information about vehicle movement and position; however, it remains an approximation of a larger physical system and can be impacted by noise, sparsity, and latency in dense urban environments. Social media data such as images and landmarks (e.g., road names) can provide additional context used for geographical estimation. The study of numerical analysis and numerical methods for partial differential equations are therefore especially significant for modeling continuous traffic flow that these alternative approaches can at best approximate. Through this program of study, I will explore how these methods provide tools for integrating noisy, discrete data into stable and scalable models that reflect traffic flow, uncertainty, and incident impact.

The study of capturing and rendering real-world scenes enables the visualization of traffic incidents by aligning images, V2X sensor data, and physical infrastructure within smart-city environments. These techniques enhance situational awareness and support scalable understanding of real-world events that impact public infrastructure.

Name: Madison Juarez

Program of Study

Listed are the courses in science and engineering, applied mathematics, and computer science that you agreed to take on your proposed Program of Study.

University: Purdue University

Course number	Course Title	Credit hours	Term and Year	Grade	Academic Level
Science & Engineering Applications					
CS50500	Distributed Systems	3S	Fall 2028		G
IE67400	Computer and Communication Methods for Production Control	3S	Fall 2027		G
Mathematics and Statistics					
CS51400	Numerical Analysis	3S	Fall 2026		G
CS61500	Numerical Methods for Partial Differential Equations I	3S	Spring 2027		G
High-Performance Computing					
CS52500	Parallel Computing	3S	Fall 2026		G
CS55100	Cloud Computing Fundamentals	3S	Fall 2026		G
Computer Science and Computer Engineering					
CS53600	Data Communication and Computer Networks	3S	Spring 2027		G
CS63500	Capturing And Rendering Real-World Scenes	3S	Fall 2027		G

I have read this program of study and affirm that, in my opinion, it satisfies the fellowship program requirements. This POS has been approved by my advisor, [REDACTED] and I understand that, if offered a fellowship, my advisor and I are required to sign this page and send it to the Krell Institute.

Student's signature _____ Date _____
Graduate Advisor: [REDACTED]
Graduate Advisor's Institute: [REDACTED]
Graduate Advisor signature _____ Date _____
Krell Institute (Office use only) _____

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